NLP Final Project Report of Part A- IMDb Movie Review Sentiment Analysis.

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BATCH: Data Science (May 15th)

NOTE: I was having a problem my loom ID, So I recorded the video on my friend's ID. Please don't get confused. I am introducing myself in the video.

Project Explanation Video: https://www.loom.com/share/bfba35f6b0964cb9a98c6e08459fcbc8

Introduction

In Part A, the objective was to build a sentiment analysis model using the IMDb movie review dataset. The goal was to classify each review as **positive** or **negative** using natural language processing and machine learning techniques.

Dataset Description

The dataset contained movie reviews along with corresponding sentiment labels (positive/negative). Each row represented a text review written by a user.

Data Preprocessing

The following preprocessing steps were applied to clean and prepare the text:

- Converted text to lowercase
- Removed HTML tags
- Removed URLs
- Removed punctuation and special characters
- Tokenized text
- Removed stopwords
- Applied Porter Stemmer
- Joined tokens back into cleaned text

A new cleaned column (`final_text`) was created.

Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) was used to convert the cleaned text into numerical vectors. Bigrams and unigrams were included, and the vocabulary size was limited using `max_features`.

Train–Test Split

The data was divided into training and testing sets using an 80:20 split.

Model Development

Three different machine learning models were trained:

- **Logistic Regression**
- **Multinomial Naive Bayes**
- **Support Vector Machine (Linear SVM)**

Cross-validation (5-fold) was applied to each model for more reliable performance.

Model Evaluation

Each model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- Cross-validation scores

Logistic Regression achieved the highest performance among the three models.

Results Comparison

The accuracy and F1-scores of all models were compared. Visualization included:

- Accuracy comparison bar chart
- Confusion matrices for each model

Based on overall performance, **Logistic Regression** was selected as the best model for sentiment classification.

Conclusion

Part A successfully demonstrated the complete workflow of text processing and sentiment analysis using the IMDb dataset. After comparing multiple models, Logistic Regression gave the best results due to its ability to handle high-dimensional sparse TF-IDF features effectively.